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ITA0412 COAT - 02

1) Campus Canteen Billing -

Pseudo Code :

```
START
INPUT  Price vector
INPUT  qty vector
FOR  each item :
        item Bill [i] = Price [i] * qty[i]
END FOR
total Bill  = SUM (item Bill)
IF  total Bill  > 500  THEN
        discount  = total Bill * 0.10
Else
        discount =0
END IF
final  Amount  =  total Bill  —  discount
DISPLAY  itemBill
DISPLAY  total Bill
DISPLAY  discount
DISPLAY  final Amount
STOP
```

2) Student Marks Analyser :

Pseudo code :

```
START
INPUT  marks  vector  of  40  student
 average  :  SUM (marks) / 40
```

highest : MAX (marks)

lowest : MIN (Marks)

fail count = 0

FOR each mark in mark

IF mark < 40 THEN

fail count = fail count + 1

END IF

END FOR

Pass count = 40 - fail count

Pass Percent = [Pass count / 40] * 100

IF Pass Percent ≥ 80 THEN

DISPLAY "Class Performance : Good"

ELSE

DISPLAY average, highest, lowest

DISPLAY fail count, Pass count

STOP

3) Hostel Room Allocation Matrix :

Pseudo code :

START

INPUT Occupancy matrix of size 4x5

for each floor i from 1 to 4

vacant [i] = 0

for each room j from 1 to 5

```
IF  Occupancy [i] [j] = 0
THEN
   Vacant [i] = Vacant [i] + 1
END IF
END FOR
DISPLAY  floor i, Vacant [i]
END FOR
DISPLAY  total Vacant = SUM (Vacant)
MAX Vacant = MAX (Vacant)
MAX Floor = POSITION of Max Vacant in Vacant
DISPLAY  total Vacant
DISPLAY  max Floor
STOP
```

Monthly Electricity Bill Calculator :

Pseudo Code :

```
START
INPUT  Units
IF  units <= 100.  THEN
    energy charge = Units * 3
ELSE IF  Units <= 200  THEN
    energy charge = (100 * 3) + ((Units - 100) * 5)
    energy charge = (100 * 3) + (100 * 5) + ((Units - 200) * 8
END IF
    fixed charge = 50
```

total Bill = energy charge + fixed charge

DISPLAY Units

DISPLAY total Bill

STOP

5) Sales Report Using Matrix Operations

Pseudo Code :

START

INPUT Sales matrix if size 3×4

FOR each Product i from 1 to 3

Product Total [i] = sum of row i

END FOR

FOR each week j from 1 to 4

week Total [J] = SUM of column j

END FOR

best Product = POSITION of MAX (Product total)

best week = POSITION of MAX (week total)

transposed Matrix = TRANSPOSE (Sales)

DISPLAY Product total

DISPLAY week Total

DISPLAY best week

DISPLAY transpose Matrix

STOP

Library Book Management Using lists

Pseudo Code

```
START
CREATE list of 5 books
FOR each book
IF Year >2015 THEN
DISPLAY book details
 END IF
 END FOR
INPUT borrow title
    found = FALSE
FOR EACh book
IF  title = borrow title THEN
    found = TRUE
   available = FAISE
DISPLAY "Book borrowed"
BREAK
END IF
 END FOR
IF found = FAISE THEN
DISPLAY 'title not found'
END IF
STOP
```

7) Temperature Data cleaning :

Pseudo Code :

```
START
INPUT temperature
 missing count = COUNT (NA)
 mean Temp = MEAN (valid temperature)
 REPLACE each NA with mean temp
 FOR each temperature
   IF temperature < 20 THEN
     category = "Cold"
   ELSE IF temperature <= 30
   THEN
     category = "Moderate"
   ELSE
     category = "Hot"
END IF
 STORE category
 END FOR
 DISPLAY missing count, temperatures
 STOP
```

Exam Seating Using 3D Arrays :

Pseudo Code :

```
START
INPUT Seating array
FOR each block
    count =0
FOR each row & column
IF seat !=0 THEN
    count =count +1
END IF
END FOR
STORE COUNT
END FOR

DISPLAY Occupied Seats per block
DISPLAY block with MAX Occupancy
DISPLAY Seating [2] [3] [4]
STOP
```

Employee Data Frame Processing

Pseudo Code :

```
START
INPUT employee data frame
DISPLAY employee where
Department = "IT" AND Salary > 50000
FOR each department
 average = MEAN (department Salaries)
DISPLAY department, average
```

```
END FOR
FOR each employee
    IF Salary < 30000 THEN
        New Salary = Salary * 10
    ELSE
        New Salary = Salary
    END IF
END FOR
ADD New Salary Column
DISPLAY updated data frame
STOP
```

10) Prime Number & Fibonacci Series Generator;

Pseudo Code :

```
START
INPUT N
WHILE N is not a Positive Integer
INPUT N again
END WHILE
CREATE empty Prime Vector
FOR numbers from 2 to N
    is Prime = TRUE
FOR divisor from 2 to number - 1
IF number MOD division = OTHER
    is Prime = FALSE
        BREAK
Display fibonacci Vector
STOP
```